

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Line plots are useful to show large quantities of data. *True or False*
Line plots are useful to show small quantities of data.
2. Histograms and stem-and-leaf plots can show large quantities of data and the overall shape. *True or False*
Histograms and box plots can show large quantities of data and the overall shape.
3. Box plots show the overall shape and information to calculate the 5-number summary. *True or False*
4. A fuel efficiency, in miles per gallon, of 250 cars were surveyed. Which graphical display would best display the data?

- ☐ Box plot
 - ☐ Line plot
 - ☒ Histogram
 - ☐ Stem-and-leaf plot
5. Explain your reasoning for question 4.
250 is a large quantity of data that could easily be grouped into ranges, so it is best to use a histogram.
6. Greg was in the market for a new SUV. He recorded the fuel efficiency for his top 12 choices.

{22.5, 17.2, 18, 20.2, 13.6, 19.4, 14, 20.4, 11.6, 16, 15.9, 20}

Explain which display would be better for the data, a line plot or a histogram.

Since there are no repeating values, it is best for Greg to use a histogram so that he can group his data into ranges.

A survey was conducted to middle school students in order to determine their favorite type of music. The frequency and relative frequency tables below represent the data collected.

Frequency Table

Type of Music	Frequency
Rap	65
Hip hop	87
Rock	32
Other	16

Relative Frequency Table

Type of Music	Frequency
Rap	32.5%
Hip hop	43.5%
Rock	16%
Other	8%

7. In order to create a circle graph, the data from the table is needed.

☐ frequency
☒ relative frequency

8. In order to create a bar graph, the data from the table is needed.

☒ frequency
☐ relative frequency

9. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☐ Frequency Table
☒ Box plot
☒ Line Plot
☒ Histogram
☐ Circle Graph
☒ Stem-and-leaf plot
☐ Bar Graph

10. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☒ Frequency Table
☐ Box plot
☒ Line Plot
☐ Histogram
☒ Circle Graph
☐ Stem-and-leaf plot
☒ Bar Graph